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--Input Vectors are Mathematical Expression
--Automated Output Verification
LIBRARY IEEE;
USE IEEE.STD_LOGIC _1164.all;
USE IEEE.STD_LOGIC_UNSIGNED.all;
USE IEEE.STD_LOGIC_ARITH.all;
USE work.my_conversion_package .ALL;

ENTITY testadder2bit is
END testadder2bit;

ARCHITECTURE testbench OF testadder2bit IS

  COMPONENT adder2bit
  PORT (x,  y          :  IN STD_LOGIC_VECTOR(1 DOWNTO 0);
          cin          :  IN STD_LOGIC;
          sum          :  OUT STD_LOGIC_VECTOR(1 DOWNTO 0);
          cout1        :  OUT STD_LOGIC);
  END COMPONENT;

  CONSTANT  NextInputDelay          :  TIME := 20 ns;
  CONSTANT  NumOfVectors            :  INTEGER:= 32;

  SIGNAL  DesignIn                  :  STD_LOGIC_VECTOR(4 DOWNTO 0);
  SIGNAL  DesignOut                 :  STD_LOGIC_VECTOR(2 DOWNTO 0);
  SIGNAL  DesignOutExpected         :  STD_LOGIC_VECTOR(2 DOWNTO 0);

BEGIN

--Design under test instantiation
  u1: adder2bit PORT MAP( x => DesignIn(2 DOWNTO 1), y => DesignIn(4 DOWNTO 3),
                          cin => DesignIn(0), sum => DesignOut(1 DOWNTO 0), cout1 => DesignOut(2) );

  Stimuli: PROCESS
    VARIABLE DesignOutExpected : STD_LOGIC_VECTOR(2 DOWNTO 0);
    VARIABLE DesignOutExpectedInt : INTEGER RANGE 0 TO 7;
  BEGIN
    FOR i in 0 TO NumOfVectors-1 LOOP -- check all combinations
        DesignIn <= CONV_STD_LOGIC_VECTOR(i, 5); -- stimulus
        WAIT FOR NextInputDelay;
 -- DesignOutExpectedInt  =  x + y + cin
        DesignOutExpectedInt  :=  CONV_INTEGER(DesignIn(4 DOWNTO 3))+
                                  CONV_INTEGER(DesignIn(2 DOWNTO 1)) +  CONV_INTEGER(DesignIn(0));
        DesignOutExpected := CONV_STD_LOGIC_VECTOR(DesignOutExpectedInt,3);
        ASSERT DesignOut = DesignOutExpected
          REPORT "Error: input = " & tostring(DesignIn)& " Expected Output = " & tostring(DesignOutExpected)
          SEVERITY error;
    END LOOP;
    REPORT "Simulation Done!";
    WAIT; -- end simulation
  END PROCESS;

END testbench;
```